

Diagnostic (for printing)

Wednesday, 23 August 2023

4:19 pm



KEY

MATH 101 CALCULUS 1 DIAGNOSTIC TEST FALL 2023

NAME:

HU ID:

SECTION:

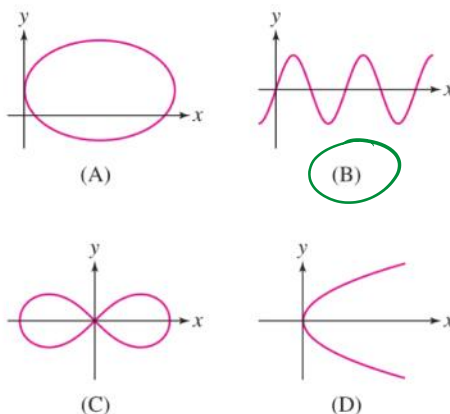
INSTRUCTOR:

Duration: 50 minutes

INSTRUCTIONS

- This test has **23 questions** worth a total of **60 marks**. There are **12 printed pages** (please count these now!)
- Attempt **all** questions.
- Show **all** relevant working in the space next to each question. Generally, correct answers with no working seen where necessary will score a **zero**.
- For any MCQs, clearly **circle only ONE** final answer.
- MCQs will be marked as follows:
 - +2 for a correct answer with justification/working
 - -1 for an incorrect answer
 - 0 if no attempt
 - 0 if correct answer with no relevant working leading to the correctly marked answer

Q. 1 Which of the curves below is the graph of a function (the independent variable is x)? [2]



Q. 2 Consider $w > 1$. Arrange the following expressions in order, from smallest to largest: [2]

$$w^0 \quad \frac{w^5}{w^6} \quad w^{-2} \quad w^5$$

$$\underline{w^{-2}, w^{-1}, w^0, w^5}$$



Q. 3 Solve the equation $\log(x^5) = (\log x)^2$ for x .

[3]

$$\begin{aligned} \Rightarrow 5 \log(x) - (\log(x))^2 &= 0 \\ \Rightarrow \log(x) (5 - \log(x)) &= 0 \\ \Rightarrow \log(x) = 0 \quad \vee \quad 5 - \log(x) &= 0 \\ x = 1 \quad , \quad x = e^5 \end{aligned}$$

Q. 4 Let $A = \{0, 2, 3, 4, 8\}$, $B = \{0, 1, 2, 3\}$, $C = \{0, 1, 5, 6, 7, 8\}$. Using proper set notation, list the elements of the following set: [2]

$$(B \cup A) \cap C = \{0, 1, 8\}$$

Q. 5 What relationship is represented in the table below between the pressure P and the temperature T of a fluid? [2]

Temperature ($^{\circ}\text{C}$)	Pressure (kPa)
40	1365.8
45	1385.40
55	1424.60
70	1483.40
80	1522.60

- A. P varies linearly with T .
- B. P is inversely proportional to T .
- C. P is an increasing and quadratic function of T .
- D. P is proportional to T .



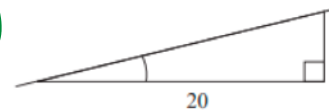
Q. 6 $f(x) = x^2$ is invertible over:

[2]

- A. $x: x \geq 0$.
- B. $x: x \neq 0$.
- C. $x: |x| > 0$.
- D. $x: |x| \geq 0$.

Q.7 Ramp structures help people going into a building. For the ramp given below, what angle gives the recommended gradient? [2]

$$\theta = \tan^{-1} \left(\frac{P}{B} \right) = \tan^{-1} \left(\frac{1}{20} \right)$$



NOT TO SCALE

$$= 0.4996 \text{ radian or } 2.86 \text{ Degree}$$

Q. 8 Find the domain of the function $f(x) = \log(2 - 3x)$, giving your answer using *interval notation*. [2]

$$2 - 3x > 0$$

$$x < \frac{2}{3} \text{ OR } x \in (-\infty, \frac{2}{3})$$

Q. 9 Let $f(x) = \sqrt{x}$ and $g(x) = \frac{1}{1+x^2}$. Write $f \circ g$ in the most simplified form. [2]

$$f \circ g = f(g(x)) \Rightarrow f\left(\frac{1}{1+x^2}\right) = \sqrt{\frac{1}{1+x^2}}$$

OR

$$\frac{1}{\sqrt{1+x^2}}$$

3



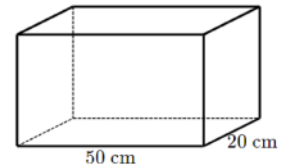
Q. 10 Zach keeps his pet chameleon Pinky in a cuboidal container with the dimensions shown on the right. There is sand at the bottom of the container that reaches a height of 8 centimeters.

He gets a new container for Pinky that is larger. The base of this new container is 64 by 25 centimeters. Zach moves the existing sand to the new container. How deep will the sand be in the new container? [2]

- A. 6 cm
- ☒ B. 5 cm
- C. 4 cm
- D. 3 cm

$$50 \times 20 \times 8 = 64 \times 25 \times h$$

$$h = 5$$



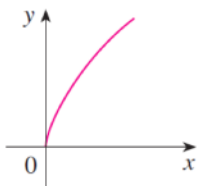
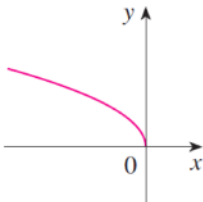
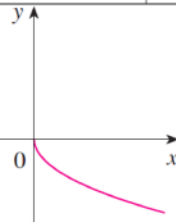
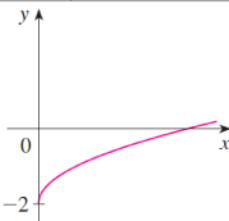
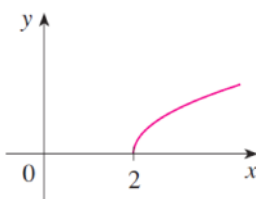
Q. 11 Simplify the following in a factor form: $\frac{6x^2+9x}{x^2+8x+15} \div \frac{4x+6}{x^2-9}$ [3]

$$\frac{6x^2+9x}{x^2+8x+15} \times \frac{x^2-9}{4x+6} \Rightarrow \frac{3x \cancel{(2x+3)}}{(x+3)\cancel{(x+5)}} \cdot \frac{(x-3)\cancel{(x+3)}}{4\cancel{(2x+3)}}$$

$$\Rightarrow \frac{3x(x-3)}{2(x+5)}$$



Q. 12 Consider the graph of $y = \sqrt{x}$. Match the transformed graphs with the correct equations:

1		A	$y = \sqrt{-x}$
2		B	$y = \sqrt{x} - 2$
3		C	$y = -\sqrt{x}$
4		D	$y = 2\sqrt{x}$
5		E	$y = \sqrt{x-2}$

(1) D (4) B
 (2) A (5) E
 (3) C [2.5]



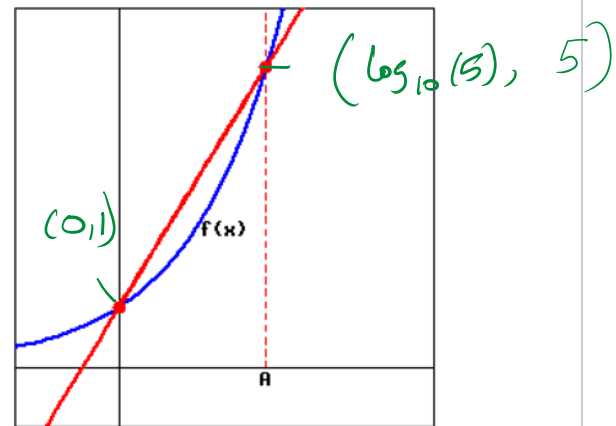
Q. 13 Refer to the graph on the right. Find the equation of the straight line illustrated in the figure, given that $f(x) = 10^x$, and the point A is at $x = \log_{10}(5)$. [3]

At $x=0$, $y = f(0) = 1$

$$m = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow \frac{4}{\log_{10}(5)}$$

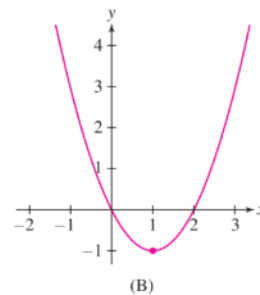
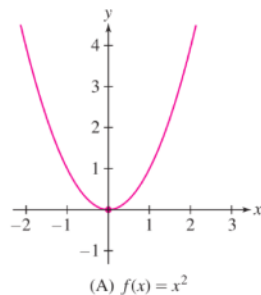
$y = mx + c$, Since c is 1
At $x=0$, $y=1$

$y = \frac{4}{\log_{10}(5)} x + 1$ or $y = 5.73 x + 1$



$\therefore \log_{10}(5) = 0.698$

Q. 14 Graph A below shows $f(x) = x^2$. The equation of graph B is: $(x-1)^2 - 1$ [2]





Q. 15 Briefly explain why the following statement is True or False:

[2.5]

If $b < 3$, then $|b| < 3$.

False, let $b = -5 \therefore b < 3$ but
 $|b| = 5 > 3$.

Q. 16 What are the domain and inverse of $y = f(x) = \frac{3x+2}{5x-1}$?

[2]

- A. Domain is $x = \frac{1}{5}$. Inverse: $x = \frac{5y-1}{3y+2}$
- B. Domain is $\{x: x \neq -\frac{2}{3}\}$. Inverse: does not exist
- ☒ C. Domain is $\{x: x \neq \frac{1}{5}\}$. Inverse: $x = \frac{y+2}{5y-3}$
- D. Domain is $x = -\frac{1}{5}$. Inverse: does not exist

Q. 17 Bacteria X has a growth rate of 160% per hour. Some amount of bacteria X are accidentally introduced into some egg salad. Three hours after contamination, there are 20,000 bacteria X in the salad. Find the initial number of bacteria X that was introduced into the salad.

[3]

$$\begin{aligned} \text{Pop.}(t) &= A \times (\text{Per centy})^t & \therefore t = \text{time} \\ & & A = \text{initial Pop.} \\ 20,000 &= A \times (1.6)^3 & \text{at } t=3 \\ A &= 4882 \end{aligned}$$



Q. 18 Use trigonometric identities to solve $\sin(2\theta) - \cos(2\theta) = 0$, for $0 \leq \theta < 2\pi$.
Final answer(s) must be in terms of π .

[4]

$$\sin(2\theta) - \cos(2\theta) = 0$$

$$\sin(2\theta) = \cos(2\theta) \Rightarrow 2\theta = \frac{\pi}{4} \quad \therefore \begin{matrix} \cos \theta = \sin(\theta) \\ \text{when} \\ \theta = \pi/4 \end{matrix}$$

$\Rightarrow \theta = \frac{\pi}{8}$, It would be periodic by $\pi/2$

So, final value would be $\frac{\pi}{8}, \frac{\pi}{8} + \frac{\pi}{2}, \frac{\pi}{8} + 2\frac{\pi}{2}, \frac{\pi}{8} + 3\frac{\pi}{2}$
 $\frac{\pi}{8} + 4\frac{\pi}{2}, \frac{\pi}{8} + 5\frac{\pi}{2}$ in $0 \leq \theta \leq 2\pi$

Another sol:

$$\frac{\sin 2\theta}{\cos(2\theta)} - 1 = 0 \Rightarrow \tan(2\theta) - 1 = 0, \tan(2\theta) = 1$$

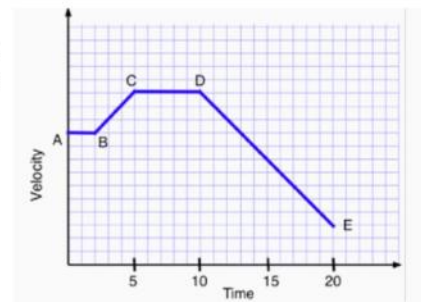
$$\Rightarrow 2\theta = \frac{\pi}{4} \Rightarrow \theta = \frac{\pi}{8}$$

same description...

Q. 19 Study the velocity-time graph for a motor car travelling on a motorway. During what stage of the journey is it accelerating at a steady rate?

[2]

- A. C to D only $\rightarrow 0 \rightarrow \frac{0}{5} \quad -10$
 B. D to E $\rightarrow 1 \rightarrow \frac{1}{5} \quad -10$
 C. A to B and C to D $\rightarrow 0$
 D. B to C $\rightarrow 1 \rightarrow \frac{3}{3} \quad \frac{0}{2}$





Q. 20 Consider the quartic function $P(x) = -2x^3 + 14x + 6$.

(a) Answer the following questions regarding end-behaviour of $P(x)$:

[2]

(i) $P(x) \rightarrow \infty$ if $x \rightarrow -\infty$

(ii) $P(x) \rightarrow -\infty$ if $x \rightarrow \infty$

(b) How many times do you expect the graph of $P(x)$ to turn? 2

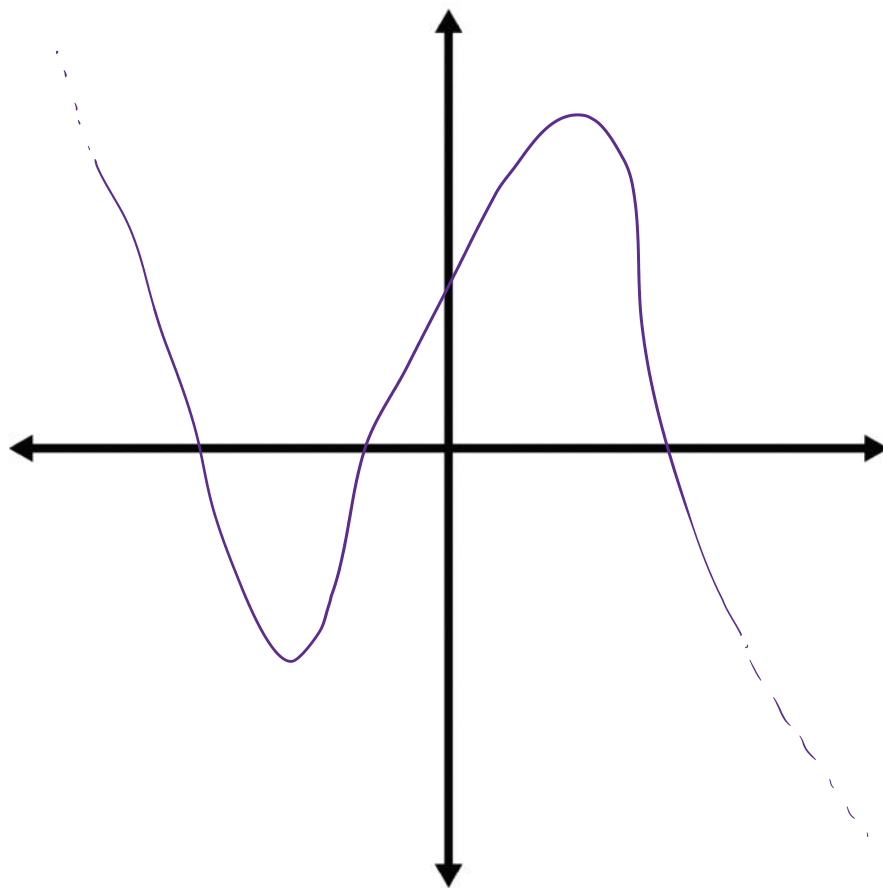
[0.5]

(c) How many roots do you expect the function to have? 3

[0.5]

(d) Hence or otherwise, **sketch** (do not plot) the graph of $P(x)$.

[2]





Q. 21 The length of a rectangle is 5 cm less than twice its width. If the perimeter of the rectangle is no more than 70 cm, what are the maximum possible dimensions of the rectangle? [3]

$$2(2w - 5) + 2w \leq 70$$

$$4w - 10 + 2w \leq 70$$

$$6w \leq 80$$

$$w \leq \frac{40}{3} \quad \& \quad l = \frac{65}{3}$$

Q. 22 John bought some pens and pencils at a bookstore. If 5 pens and 3 pencils cost \$8.40 while 5 pencils and 3 pens cost \$6 instead, find the cost of a pen and pencil. [3]

Let $x = \text{pen}$ & $y = \text{pencil}$

$$5x + 3y = 8.4 \quad \& \quad 3x + 5y = 6$$

$$x = \frac{11}{8}, \quad y = \frac{3}{8}$$

OR

$$x = 1.3, \quad y = 0.3$$



Q. 23 Given that $f(x) = \frac{2}{(x-3)^2}$, answer the questions that follow. In case of multiple answers, separate these with commas. If an asymptote or intercept does not exist, write **none**. [4]

(a) Horizontal Asymptote(s): $y = 0$

(b) Hole(s): at $x=3$ as $(3, \infty)$ (write answer(s) as ordered pairs)

(c) Vertical Asymptote(s): $x = 3$

(d) x-intercept(s): No x-intercept or at infinity $(-\infty, 0)$ & $(\infty, 0)$ (write answer(s) as ordered pairs)

(e) y-intercept(s): $(0, 2/9)$ (write answer(s) as ordered pairs)

(f) Hence, decide which of the following graphs represents $f(x)$, stating the letter: _____

